

IAA - Deliverable 2

Sueca Game Player

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March 2026

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1 Introduction

As a quick reintroduction, our project aims to develop a ML model capable of autonomously play the card game "Sueca". The basic idea is that it is trained in hopes to become good at the decision-making that occurs at a player's turn.

2 Data Processing Methodology

In order to start generating and preparing the data, we must first define **what that data should look like** first. Because it is generated synthetically, besides models used, it is also our responsibility to handle the data that is used. For that reason, we have defined some specific points.

2.1 Game State Format (Can be different from the shown)

To help the model contextualize itself in the game, we have features that track the current round of the game, alongside the overall information the player took into account before making a decision. The parts of this format that are most important are as follows:

- **Timestamps:**

The timestamps were not present in the actual game's state and structure and are being generated and stored during the data collection process, making them not as important in the game's internal system.

- **Round management:**

Each round is represented by a number from 1 to 10.

- **Cards in Hand:**

Each card usually has either a number from 2 to 7, A, K, Q, and J, as well as one of four suits: hearts, spades, clubs and diamonds. To simplify them, we gave each card a value from 0 to 39 (the number of cards in a deck). During any moment, the hand of the player is represented by an array of numbers, where each represents one of the 40 cards.

- **Cards in trick (Played Cards):**

The cards that have been played in the current trick are being stored inside a list that contains the name of the player who played the card and the number associated with said card. This list then is stored inside another list that contains this information for each card that was played before the current play, i.e. each player stores a list of lists with the names and cards played of everyone who played before it.

- **Player position:**

Each position is being stored as a string ('SOUTH', 'NORTH', 'EAST', 'WEST'). These strings are defined inside a positions class using ENUM logic.

- **Card suits:**

Each card suit is being stored as its respective utf-8 character.

- **Game Conditions:**

The game is being managed by a server and the communication between the server and each individual player is being managed by a middleware system that makes sure the information flows properly from one side to the other. This format makes sure that some part of the information is truly secret and the server only receives what it needs to run the game and manage all of the actions properly.

It is because of this methodology that we have determined that, in case of underwhelming results from our chosen models later on, we may have to keep iterating and improving the quality of our defined dataset.

2.2 The data gathering process

• Output collection

After deciding what that data should look like and what information the model would need to learn to play properly without extreme bias develop any bad habits easily, we have decided to build a python script that would listen in on the server where the game is being played and took information from the AI players' logs and state and record that information in a csv file. The script also provides some additional helpful information that might or might not be used later, taking into account the possibility of the model creating hidden bias from said information.

• Example output image

```
1 timestamp,round_number,player,position,hand_before,legal_moves,chosen_card,cards_in_trick,position_in_trick,lead_suit,trump
2 2026-04-23T17:26:24.248659+00:00,1,S4,SOUTH,"[3, 4, 11, 14, 21, 29, 31, 34, 35, 36]", "[3, 4, 11, 14, 21, 29, 31, 34, 35, 36]",3,[],0,♥
3 2026-04-23T17:26:24.306769+00:00,1,E4,EAST,"[5, 7, 17, 23, 25, 27, 30, 33, 37, 38]", "[5, 7]",7,"[['S4', 3]]",1,♠♥
4 2026-04-23T17:26:24.358714+00:00,1,N4,NORTH,"[6, 9, 12, 13, 19, 20, 22, 26, 32, 39]", "[6, 9]",9,"[['S4', 3], ['E4', 7]]",2,♠♥
5 2026-04-23T17:26:24.437588+00:00,1,W4,WEST,"[0, 1, 2, 8, 10, 15, 16, 18, 24, 28]", "[0, 1, 2, 8]",2,"[['S4', 3], ['E4', 7], ['N4', 9]]",3,♠♥
6 2026-04-23T17:26:26.166294+00:00,2,N4,NORTH,"[6, 12, 13, 19, 20, 22, 26, 32, 39]", "[6, 12, 13, 19, 20, 22, 26, 32, 39]",12,[],0,♥
7 2026-04-23T17:26:26.304908+00:00,2,W4,WEST,"[0, 1, 8, 10, 15, 16, 18, 24, 28]", "[10, 15, 16, 18]",15,"[['N4', 12]]",1,♠♥
8 2026-04-23T17:26:26.358674+00:00,2,S4,SOUTH,"[4, 11, 14, 21, 29, 31, 34, 35, 36]", "[11, 14]",11,"[['N4', 12], ['W4', 15]]",2,♠♥
9 2026-04-23T17:26:26.416572+00:00,2,E4,EAST,"[5, 17, 23, 25, 27, 30, 33, 37, 38]", "[17, 17]",17,"[['N4', 12], ['W4', 15], ['S4', 11]]",3,♠♥
10 2026-04-23T17:26:28.209538+00:00,3,E4,EAST,"[5, 23, 25, 27, 30, 33, 37, 38]", "[5, 23, 25, 27, 30, 33, 37, 38]",33,[],0,♥
11 2026-04-23T17:26:28.340877+00:00,3,N4,NORTH,"[6, 13, 19, 20, 22, 26, 32, 39]", "[32, 39]",32,"[['E4', 33]]",1,♠♥
12 2026-04-23T17:26:28.429204+00:00,3,W4,WEST,"[0, 1, 8, 10, 16, 18, 24, 28]", "[0, 1, 8, 10, 16, 18, 24, 28]",10,"[['E4', 33], ['N4', 32]]",2,♠♥
13 2026-04-23T17:26:28.488005+00:00,3,S4,SOUTH,"[4, 14, 21, 29, 31, 34, 35, 36]", "[31, 34, 35, 36]",35,"[['E4', 33], ['N4', 32], ['W4', 10]]",3,♠♥
14 2026-04-23T17:26:30.286048+00:00,4,S4,SOUTH,"[4, 14, 21, 29, 31, 34, 36]", "[4, 14, 21, 29, 31, 34, 36]",4,[],0,♥
15 2026-04-23T17:26:30.337300+00:00,4,E4,EAST,"[5, 23, 25, 27, 30, 37, 38]", "[5, 5]",5,"[['S4', 4]]",1,♠♥
16 2026-04-23T17:26:30.480595+00:00,4,N4,NORTH,"[6, 13, 19, 20, 22, 26, 39]", "[6, 6]",6,"[['S4', 4], ['E4', 5]]",2,♠♥
17 2026-04-23T17:26:30.611275+00:00,4,W4,WEST,"[0, 1, 8, 16, 18, 24, 28]", "[0, 1, 8]",8,"[['S4', 4], ['E4', 5], ['N4', 6]]",3,♠♥
18 2026-04-23T17:26:32.455925+00:00,5,N4,NORTH,"[13, 19, 20, 22, 26, 39]", "[13, 19, 20, 22, 26, 39]",22,[],0,♥
19 2026-04-23T17:26:32.580760+00:00,5,W4,WEST,"[1, 8, 16, 18, 24, 28]", "[24, 28]",24,"[['N4', 22]]",1,♠♥
20 2026-04-23T17:26:32.630499+00:00,5,S4,SOUTH,"[14, 21, 29, 31, 34, 36]", "[21, 29]",29,"[['N4', 22], ['W4', 24]]",2,♥♥
21 2026-04-23T17:26:32.719326+00:00,5,E4,EAST,"[23, 25, 27, 30, 37, 38]", "[23, 25, 27]",27,"[['N4', 22], ['W4', 24], ['S4', 29]]",3,♥♥
22 2026-04-23T17:26:34.527018+00:00,6,S4,SOUTH,"[14, 21, 31, 34, 36]", "[14, 21, 31, 34, 36]",31,[],0,♥
23 2026-04-23T17:26:34.631146+00:00,6,E4,EAST,"[23, 25, 30, 37, 38]", "[30, 37, 38]",38,"[['S4', 31]]",1,♠♥
24 2026-04-23T17:26:34.701759+00:00,6,N4,NORTH,"[13, 19, 20, 26, 39]", "[39]",39,"[['S4', 31], ['E4', 38]]",2,♠♥
25 2026-04-23T17:26:34.851147+00:00,6,W4,WEST,"[1, 8, 16, 18, 28]", "[1, 8, 16, 18, 28]",8,"[['S4', 31], ['E4', 38], ['N4', 39]]",3,♠♥
26 2026-04-23T17:26:36.611207+00:00,7,N4,NORTH,"[13, 19, 20, 26]", "[13, 19, 20, 26]",19,[],0,♥
27 2026-04-23T17:26:36.650355+00:00,7,W4,WEST,"[1, 16, 18, 28]", "[16, 18]",18,"[['N4', 19]]",1,♠♥
28 2026-04-23T17:26:36.704417+00:00,7,S4,SOUTH,"[14, 21, 34, 36]", "[14, 14]",14,"[['N4', 19], ['W4', 18]]",2,♠♥
29 2026-04-23T17:26:36.768760+00:00,7,E4,EAST,"[23, 25, 30, 37]", "[23, 25, 30, 37]",25,"[['N4', 19], ['W4', 18], ['S4', 14]]",3,♠♥
30 2026-04-23T17:26:38.516226+00:00,8,E4,EAST,"[23, 30, 37]", "[23, 30, 37]",23,[],0,♥
31 2026-04-23T17:26:38.630133+00:00,8,N4,NORTH,"[13, 20, 26]", "[20, 26]",26,"[['E4', 23]]",1,♠♥
32 2026-04-23T17:26:38.669092+00:00,8,W4,WEST,"[1, 16, 28]", "[28, 28]",28,"[['E4', 23], ['N4', 26]]",2,♥♥
33 2026-04-23T17:26:38.732209+00:00,8,S4,SOUTH,"[21, 34, 36]", "[21, 21]",21,"[['E4', 23], ['N4', 26], ['W4', 28]]",3,♥♥
34 2026-04-23T17:26:40.532123+00:00,9,W4,WEST,"[1, 16]", "[1, 16]",1,[],0,♥
35 2026-04-23T17:26:40.560179+00:00,9,S4,SOUTH,"[34, 36]", "[34, 36]",36,"[['W4', 1]]",1,♠♥
36 2026-04-23T17:26:40.609789+00:00,9,E4,EAST,"[30, 37]", "[30, 37]",37,"[['W4', 1], ['S4', 36]]",2,♠♥
37 2026-04-23T17:26:40.759597+00:00,9,N4,NORTH,"[13, 20]", "[13, 20]",20,"[['W4', 1], ['S4', 36], ['E4', 37]]",3,♠♥
38 2026-04-23T17:26:42.521076+00:00,10,N4,NORTH,[13],[13],13,[],0,♥
39 2026-04-23T17:26:42.578201+00:00,10,W4,WEST,[16],[16],16,"[['N4', 13]]",1,♠♥
40 2026-04-23T17:26:42.629112+00:00,10,S4,SOUTH,[34],[34],34,"[['N4', 13], ['W4', 16]]",2,♠♥
41 2026-04-23T17:26:42.682482+00:00,10,E4,EAST,[30],[30],30,"[['N4', 13], ['W4', 16], ['S4', 34]]",3,♠♥
```

• Data treatment

As shown in the image above, the information being gathered is currently being stored by game. We plan on either adding a tag that distinguishes each game. It would be done by saving each game in its own separate file and merging them all later in a file that would include a unique ID for each different game. A simple python script will likely be the tool used for the merging process.

To try to prevent hidden bias as well as possible, some of the data that is being collected is going to be processed before applying any further operations. This data includes, but is not limited to:

- Removing the timestamp
- Making the suit icons a number between 0 and 3
- Making the positions a number between 0 and 3
- Normalizing positions by associating a player to their position and treating them as the same thing

These decisions are not final and their execution may not be present in the final version

- **Setbacks during data gathering**

During data gathering, we had multiple setbacks that complicated the development of the script which handpicked all the information and stored it for later use, especially in the process of trying to figure out **how** and **where** the script should get that information from.

It started from getting all of it from the server’s game state, which turned out fine for some time, however, the server does not keep track of each player’s individual hand and their legal plays to prevent information leaks and make sure that each player’s hand is truly secret. To solve this issue, we had to make the script also take some information directly from the players, that being their hand before the play was made, the legal plays that the player could do, the card that was played and the cards that were played before them in that same trick (along with who played those cards).

With that adaptation, we also had to swap the information flow and organization to store four entries per trick (one per play) instead of one entry with the summary of each trick, so the script didn’t break and the information could be stored properly and in a format that the model could later use well.

Currently, as shown earlier in an image of the output of a full game, the information that is being stored is mostly being taken straight from the player itself, since the information that the server provided is also present in the player’s state, making it inefficient to look in two different places for the same exact information.

3 Models Explored

In order to decide which learning approach is most suitable for this project, we considered several supervised classification models with different levels of complexity and interpretability. Since the dataset is structured as tabular game-state information, the selected models must be able to handle heterogeneous input features and support multiclass prediction.

3.1 Multiclass Logistic Regression

Multiclass logistic regression was considered as a baseline model due to its simplicity, efficiency, and interpretability. It provides a clear starting point for the analysis, especially in a problem where the target is the selection of one action among several possible legal moves. Although it is limited in its ability to capture complex non-linear relationships, it is useful as a reference model against which more advanced approaches can be compared.

3.2 Gradient Boosting

Gradient Boosting was considered as a stronger non-linear baseline. Unlike logistic regression, this method can model more complex relationships between the input variables and the target by combining multiple weak learners in a sequential way. This makes it particularly suitable for tabular datasets where the prediction may depend on subtle interactions between features such as the current trick, the trump suit, the cards already played, and the player’s position. At the same time, Gradient Boosting remains relatively accessible to interpret and evaluate compared with more complex models.

3.3 Extreme Gradient Boosting (XGBoost)

XGBoost was selected as the main candidate model for the project. It is a gradient boosting method that is particularly effective on structured tabular data and is known for its strong predictive performance. In the context of Sueca, where the relationship between the game state and the best move may be highly non-linear, XGBoost is a promising option because it can model complex interactions between features while maintaining strong generalization capabilities. Compared with simpler models, it is more computationally demanding, but it offers a better trade-off between accuracy and flexibility.

3.4 Justification of the Final Choice

After considering these models, our strategy is to use logistic regression as a baseline, Gradient Boosting as an intermediate non-linear alternative, and XGBoost as the main model to be explored further. This combination allows us to compare a simple interpretable model with two more powerful ensemble methods and to justify the final selection based on both performance and modelling suitability. Since the project involves synthetic gameplay data and a multiclass prediction task, tree-based boosting methods appear particularly well aligned with the structure of the problem.

4 A Few Ethical Considerations

Due to the nature of the project at hand, there are a few considerations that we have had to ponder and come to terms with:

4.1 Gameplay Integrity and Fairness:

As an AI model, this "player" would logically have perfect memory and therefore an advantage against other player. Besides that aspect, because of an expected ability to play at an near perfect level of rational thinking, it could be used as a tool for cheating, preventing fair play in recreational games at best, and cash-prize competitions at worst.

4.2 Transparency

Similar to the previous point, players not knowing that they are playing against an advanced intelligence could cause trouble. Therefore there is a clear issue with the fact that even though it would be playing the mathematically best moves possible every time (hopefully), in a card game like this, it may be hard for the average player to identify if they are playing against an AI by their playstyle.

4.3 Data and Privacy (In the future)

While not a current problem, due to the way in which we chose to generate our data, in the future, to better "widen" its capabilities, we may have to rely on real player data, preferably from a wide variety of players. This then creates an issue with the privacy and ownership of said data, as the people from which the data is taken from have the right to refuse consent of use.

5 Fundamental Adjustments

There are a few points that, upon further judgement, we have either decided to reevaluate, or that we failed to take into account in the first place. Its because of these, that our approach to this problem has changed slightly.

1. The Rules of the Game

One aspect we failed to mention in the previous report, is that in this game not any card can be played every time. During a certain trick, only a few cards are actually legal to play. We believed that the AI would naturally be able to understand this aspect, but obviously it cannot know the fundamental rules of the game (similar to how many points a card is worth), and so we have determined that that restriction should be addressed (to prevent the model from learning to play illegal moves).

2. The Scope of the Data

Before, we established that data would be generated synthetically, we underestimate the rate of generation that we would be capable to achieve. During testing we quickly noticed that considering the volume of the data that we would need was above what was attainable, we decided to evenly split the data generation step of this problem into the regular quality assurance part, where we ensure the features and values obtained are satisfactory, and a

performance part, where focus is placed on optimizing the processing speed of the coded game state, in order to allow faster plays, faster games, and faster results.

3. Machine Learning Problem Assertion

During the definition of the ML Problem, we mentioned that we were considering shifting the scope of the project from a supervised classification problem to one of reinforcement learning. However, due to the before mentioned troubles and shift in focus to a more Data Centric methodology, we have decided to scrap the idea of applying reinforcement learning to this project, and leave it for some context outside the current semester, as it still has potential (we just lack the time).